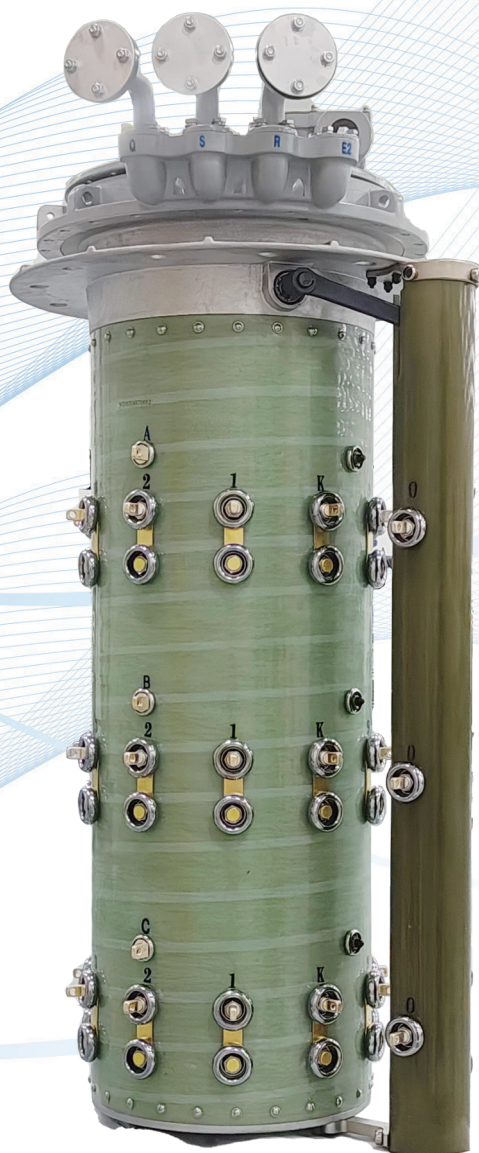




CV2 VACUUM ON-LOAD TAP-CHANGER



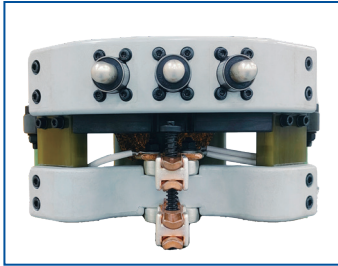
Quality Creates Value

Shanghai Huaming Power Equipment Co., Ltd.

PRODUCT CHARACTERISTICS

The CV2 vacuum on-load tap-changer is an improved combination product suitable for oil immersed transformers of highest equipment voltage of 40.5 ~ 145kV and max. rated through current 600A, complying with GB/T10230.1 and IEC60214-1 Tap changers Part 1: ***Performance requirement and test methods.***

Main contacts adopt multi-point contact, which effectively increases current carrying capability.



The vacuum interrupter is connected in series with the mechanical change-over contacts which can break the current (arc extinguishing) within a certain number of times in case of vacuum interrupter failure.

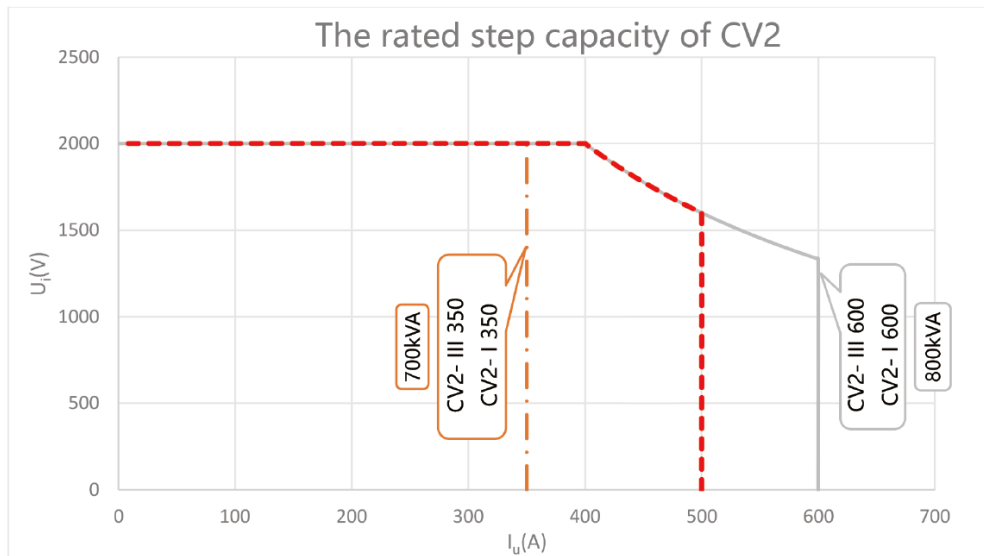


Change-over selector with three-phase vertical arrangement and high insulation margin.

Compact structure and small space occupation decrease usage cost.

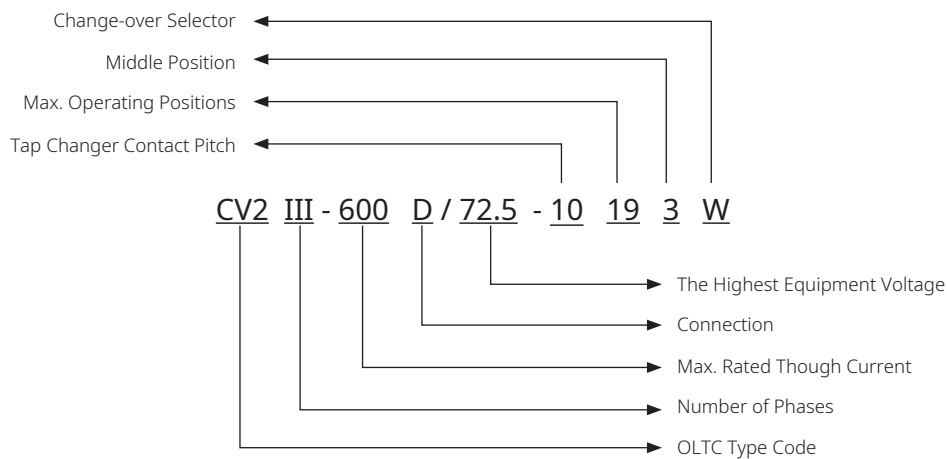
The switching circuit relies on vacuum interrupter to extinguish the arc, which can be up to 300,000 times maintenance-free, reducing the operating costs and extending the maintenance cycle of the tap changer.





$P_{StN} = I_{ux} \times U_i$ The rated step capacity is defined as the product of the step voltage and the load current, i.e. $P_s = UI$. The rated step capacity is the maximum step capacity of the on-load tap-changer under continuous operating conditions, i.e. $P_{StN} = I_{ux} \times U_i$.

Code description



Conditions

1. The operating temperature of the tap-changer in oil is $\geq -25^{\circ}\text{C}$ and $\leq 105^{\circ}\text{C}$.
2. The temperature of the ambient environment where the on-load tap-changer is to be used is -25°C to $+40^{\circ}\text{C}$.
3. The perpendicularity of the on-load tap-changer to the ground after mounting on the transformer should not exceed 2%.
4. The place of the on-load tap-changer located must be free of serious dust and other explosive or corrosive gases.

Note: For more severe environmental conditions of the on-load tap-changer or motor-drive unit, please contact with Huaming technical department for further confirmation.

Applicable Standards: IEC60214-1: 2014.

Type		CV2 350			CV2 600		
Max. Rated Through Current (A)		350			600		
Phase		I	III	III	I	III	III
Connection		-	Y	D	-	Y	D
Max. Rated Step Voltage (V)		2000 for 10 contacts 1500 for 12 contacts			2000 for 10 contacts 1500 for 12 contacts		
Rated Step Capacity (kVA)		700			800		
Short circuit capacity (kA)	Thermal(3s)	5			8		
	Dynamic (peak)	12.5			20		
Max. operating positions		12 without change-over selector; 23 with change-over selector			12 without change-over selector; 23 with change-over selector		
Insulation level to earth (kV)	The highest voltage for equipment (kV)	40.5	72.5		126	145	
	Rated separate source AC withstand voltage (kV/50Hz, 1min)	95	140		230	275	
	Rated lightning impulse withstand voltage- full wave (1.2/50μs)	250	350		550	650	
	Rated lightning impulse withstand voltage-chop wave (3~6μs)	275	285		605	715	
Mechanical life		1,500,000					
Maintenance free		300,000					
Oil compartment	Service pressure	0.03MPa					
	Leakage test	0.1MPa 24h No leakage					
	Over pressure protection	Rupture disc bursts at 0.3±20%MPa					
	Protective relay	Set oil flow speed at 1.0m/s±10%					
Motor drive unit		SHM or CMA7					